

# DIPOLLO ELETTRICO

14/10/20

①

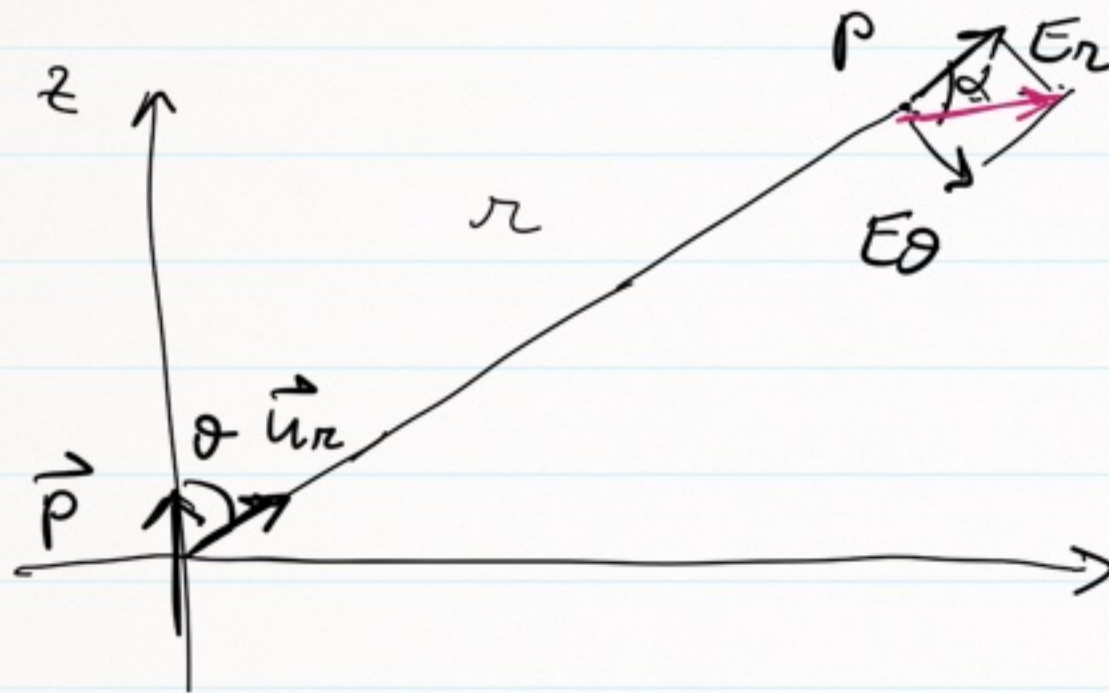
$$(a) \quad V(P) = \frac{\vec{p} \cdot \vec{u}_r}{4\pi\epsilon_0 r^2}, \quad \vec{E} = -\vec{\nabla}V$$

$$E_r = -\frac{\partial V}{\partial r} = \frac{2p \cos\theta}{4\pi\epsilon_0 r^3} = \frac{p \cos\theta}{2\pi\epsilon_0 r^3}$$

$$E_\theta = -\frac{1}{r} \cdot \frac{\partial V}{\partial \theta} = \frac{p \sin\theta}{4\pi\epsilon_0 r^3}$$

$$E_\varphi = -\frac{1}{r \sin\theta} \frac{\partial V}{\partial \varphi} = 0$$

$$\vec{E} = E_r \cdot \vec{u}_r + E_\theta \cdot \vec{u}_\theta = \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \vec{u}_r + \sin\theta \vec{u}_\theta)$$



$$|\vec{E}| = \frac{p}{4\pi\epsilon_0 r^3} \sqrt{3\cos^2\theta + 1}$$

$$\tan\alpha = \frac{E_\theta}{E_r} = \frac{1}{2} \tan\theta$$

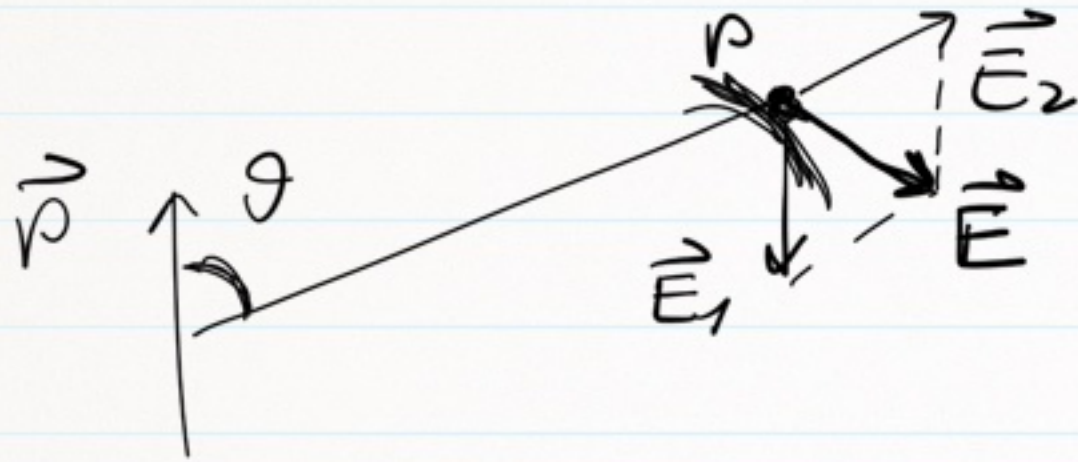


$$\vec{p} = p \cos\theta \vec{u}_r - p \sin\theta \vec{u}_\theta$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0 r^3} (3p \cos\theta \vec{u}_r - p \cos\theta \vec{u}_r + p \sin\theta \vec{u}_\theta) =$$

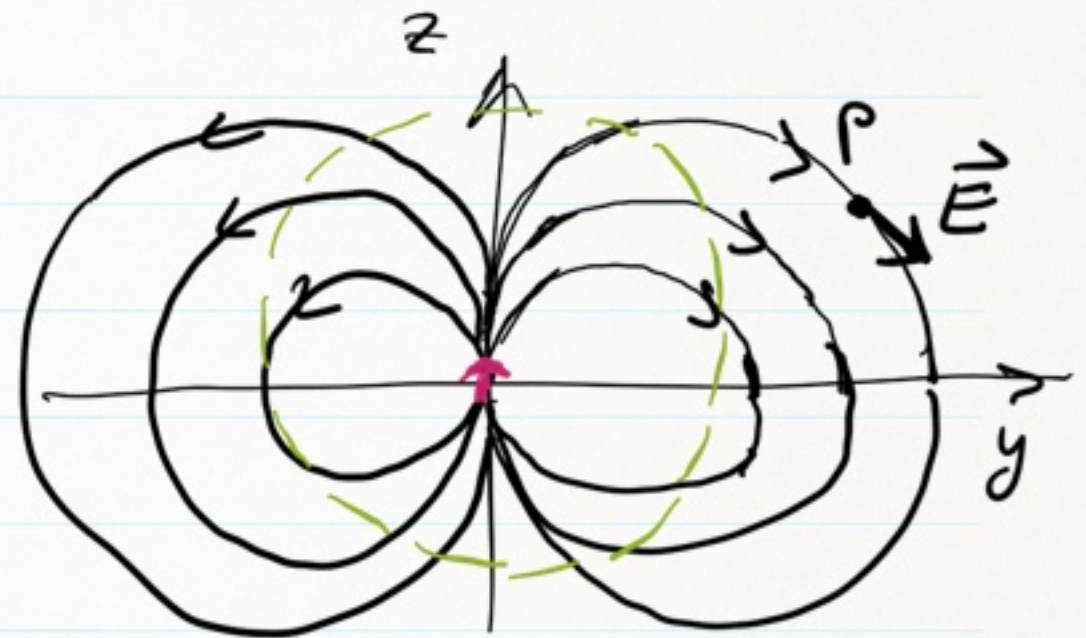
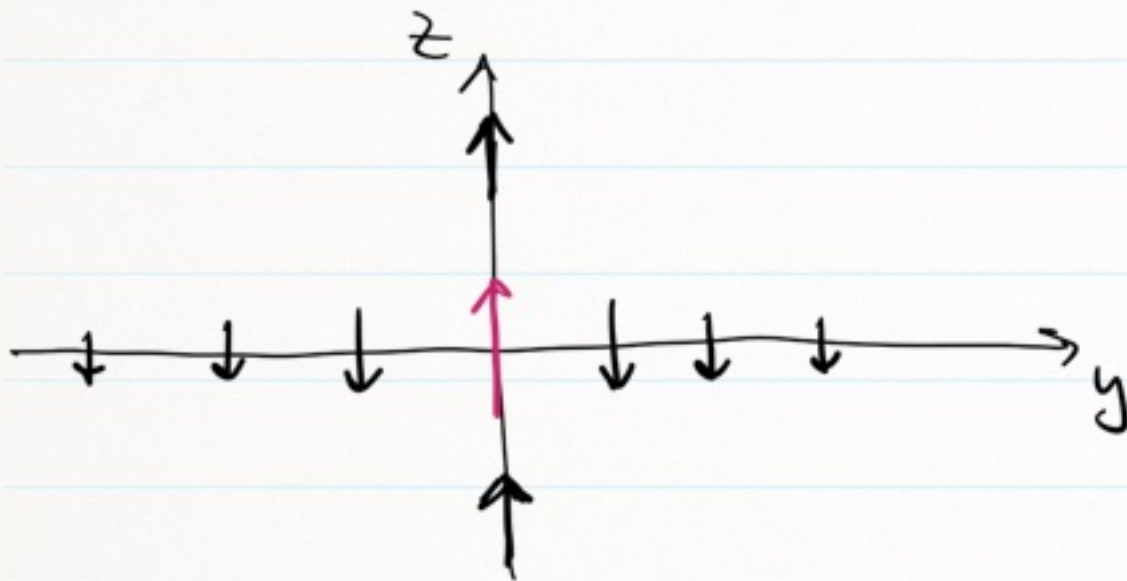
$$= \frac{1}{4\pi\epsilon_0 r^3} \left( 3 \underbrace{(\vec{p} \cdot \vec{u}_r)}_{\vec{E}_2} \vec{u}_r - \underbrace{\vec{p}}_{\vec{E}_1} \right)$$





Sull'asse del dipolo: 
$$\vec{E} = \frac{2\vec{p}}{4\pi\epsilon_0 r^3}$$

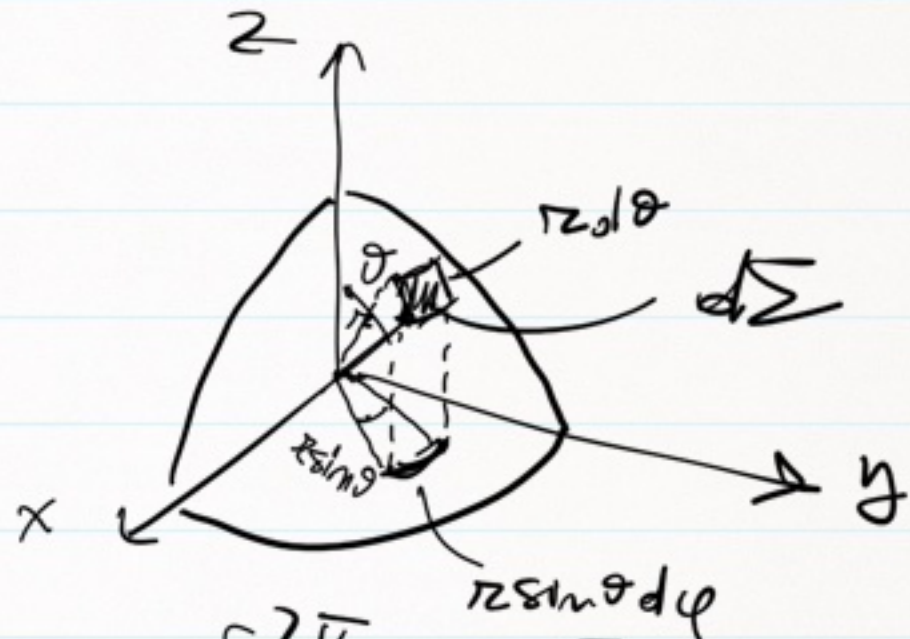
Nel piano mediano: 
$$\vec{E} = -\frac{\vec{p}}{4\pi\epsilon_0 r^3}$$



$$(c) \oint_{\Sigma} \vec{E} \cdot \vec{u}_n d\Sigma$$

$$\bullet \vec{E} = \frac{\rho}{4\pi\epsilon_0 r^3} \left( r \cos\vartheta \vec{u}_n + \sin\vartheta \vec{u}_\vartheta \right)$$

$$\bullet d\Sigma = r^2 \sin\vartheta d\vartheta d\varphi$$



$$\oint_{\Sigma} \cos\vartheta \sin\vartheta d\vartheta d\varphi = \int_0^{2\pi} d\varphi \underbrace{\int_0^{\pi} \cos\vartheta \sin\vartheta d\vartheta}_{\frac{1}{2} \sin^2\vartheta}$$

$$\frac{1}{2} [\sin^2\vartheta]_0^{\pi} = 0$$

$$\Rightarrow \oint_{\Sigma} \vec{E} \cdot \vec{u}_n d\Sigma = 0 \left( = \underbrace{\sum_i Q_{int}}_0 / \epsilon_0 \right) \text{ Gauss}$$



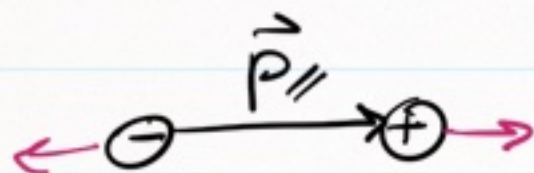
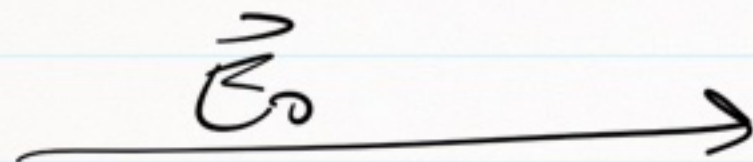
## ② Esercizio 2 (DIPLO IN CAMPO $\vec{E}_0$ UNIFORME)

$$(a) \quad U_e = q \cdot V \rightarrow U_e = qV(x+a_x, y+a_y, z+a_z) + \\ - q \rightarrow P_1(x, y, z) \quad - qV(x, y, z) \\ + q \rightarrow P_2(x+a_x, y+a_y, z+a_z)$$

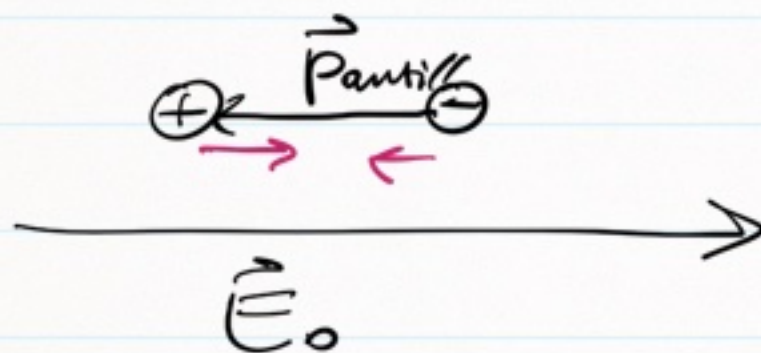
$$V(x+a_x, y+a_y, z+a_z) =$$

$$= V(x, y, z) + \underbrace{\frac{\partial V}{\partial x} \cdot a_x + \frac{\partial V}{\partial y} \cdot a_y + \frac{\partial V}{\partial z} \cdot a_z}_{}$$

$$U_e = q \left( a_x \cdot \frac{\partial V}{\partial x} + a_y \cdot \frac{\partial V}{\partial y} + a_z \cdot \frac{\partial V}{\partial z} \right) = \\ = q (\vec{a} \cdot \vec{\nabla} V) = - \vec{p} \cdot \vec{E}_0 = - p \cos \theta E_0$$

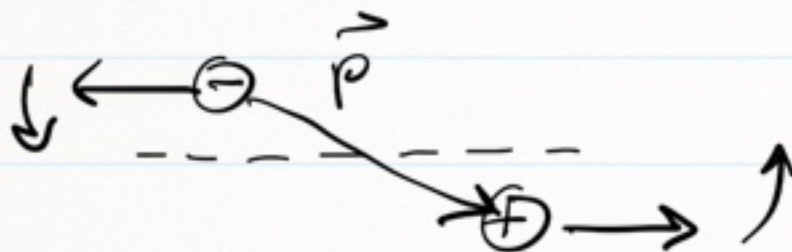


(1) STABLE @  $\theta = 0$   
 $U_e^{\min} = -pE_0$

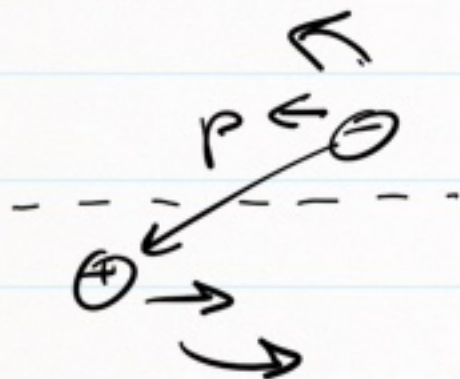


(2) UNSTABLE @  $\theta = \pi$   
 $U_e^{\max} = pE_0$

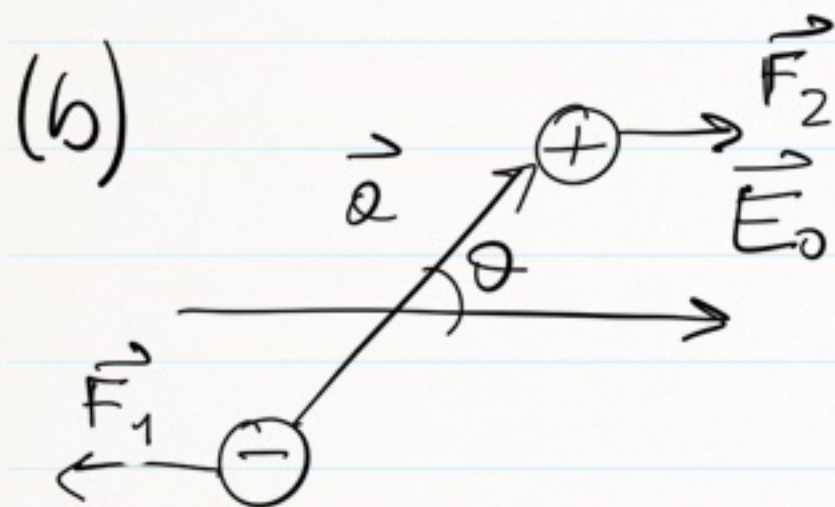
(1)



(2)







$$\begin{aligned}\vec{M} &= \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 = -\vec{r}_1 \times q\vec{E}_0 + \vec{r}_2 \times q\vec{E}_0 = \\ &= (\vec{r}_2 - \vec{r}_1) \times q\vec{E}_0 = \vec{a} \times q\vec{E}_0 = \underline{\underline{\vec{p} \times \vec{E}_0}}\end{aligned}$$

$$M = -p \sin \theta E_0 = - \underline{\underline{\frac{\partial U_e}{\partial \theta}}}$$

$$\vec{M} = -p \sin \theta E_0 \hat{u}_z$$

### ③ Esercizio 3 (Dipolo in campo non uniforme)

$$\vec{F}_1 = -q \vec{E}$$

$$\vec{F}_2 = q (\vec{E} + \Delta \vec{E}) = q \left( \vec{E} + \frac{\partial \vec{E}}{\partial x} \cdot a_x + \frac{\partial \vec{E}}{\partial y} \cdot a_y + \frac{\partial \vec{E}}{\partial z} \cdot a_z \right)$$

$$\begin{aligned} \vec{F} &= \vec{F}_1 + \vec{F}_2 = q \frac{\partial \vec{E}}{\partial x} a_x + q \frac{\partial \vec{E}}{\partial y} a_y + q \frac{\partial \vec{E}}{\partial z} a_z \\ &= p_x \cdot \frac{\partial \vec{E}}{\partial x} + p_y \cdot \frac{\partial \vec{E}}{\partial y} + p_z \cdot \frac{\partial \vec{E}}{\partial z} \end{aligned}$$

ovvero:

$$\begin{cases} F_x = p_x \cdot \frac{\partial E_x}{\partial x} + p_y \cdot \frac{\partial E_x}{\partial y} + p_z \cdot \frac{\partial E_x}{\partial z} \\ F_y = p_x \cdot \frac{\partial E_y}{\partial x} + p_y \cdot \frac{\partial E_y}{\partial y} + p_z \cdot \frac{\partial E_y}{\partial z} \\ F_z = p_x \cdot \frac{\partial E_z}{\partial x} + p_y \cdot \frac{\partial E_z}{\partial y} + p_z \cdot \frac{\partial E_z}{\partial z} \end{cases} \quad \leftarrow$$



$$\boxed{\vec{F} = (\vec{p} \cdot \vec{\nabla}) \vec{E}} \quad \leftarrow$$

$$\vec{p} \cdot \vec{\nabla} = p_x \frac{\partial}{\partial x} + p_y \frac{\partial}{\partial y} + p_z \frac{\partial}{\partial z}$$

$$(b) \quad x: \quad E_x = 0 \quad E(P) = ?$$

$$y: \quad E_y = \frac{q_0}{4\pi\epsilon_0} \cdot \frac{(y-0)}{[(y-0)^2 + (z-0)^2]^{3/2}}$$

$$z: \quad E_z = \frac{q}{4\pi\epsilon_0} \cdot \frac{z}{(y^2 + z^2)^{3/2}}$$

$$\Rightarrow \underline{F_x = 0}$$

$F_y$

$$\frac{\partial E_y}{\partial x} = 0$$

$$\frac{\partial E_y}{\partial y} = \frac{q_0}{4\pi\epsilon_0} \left[ \frac{1}{(y^2+z^2)^{3/2}} - \frac{3}{2} \frac{2yz}{(y^2+z^2)^{5/2}} \right]$$

$$= \frac{q_0}{4\pi\epsilon_0} \left[ \frac{y^2+z^2-3y^2}{(y^2+z^2)^{5/2}} \right]$$

$$= \frac{q_0}{4\pi\epsilon_0} \frac{(-2y^2+z^2)}{(y^2+z^2)^{5/2}}$$

$$\frac{\partial E_y}{\partial z} = -\frac{q_0}{4\pi\epsilon_0} \cdot \frac{3zy}{(y^2+z^2)^{5/2}}$$



(F<sub>z</sub>)

$$\frac{\partial E_z}{\partial x} = 0$$

$$\frac{\partial E_z}{\partial y} = - \frac{q_0}{4\pi\epsilon_0} \frac{3zy}{(y^2+z^2)^{5/2}}$$

$$\frac{\partial E_z}{\partial z} = \frac{q_0 (y^2 - 2z^2)}{4\pi\epsilon_0 (y^2+z^2)^{5/2}}$$

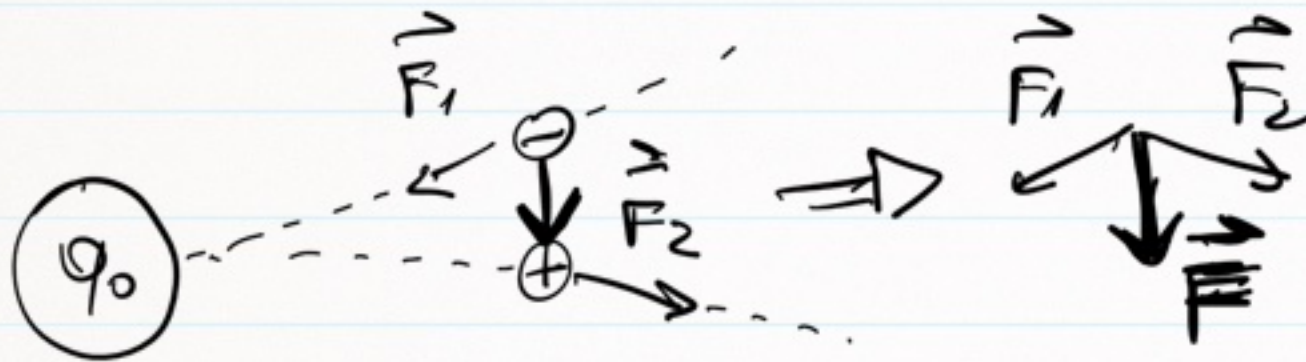
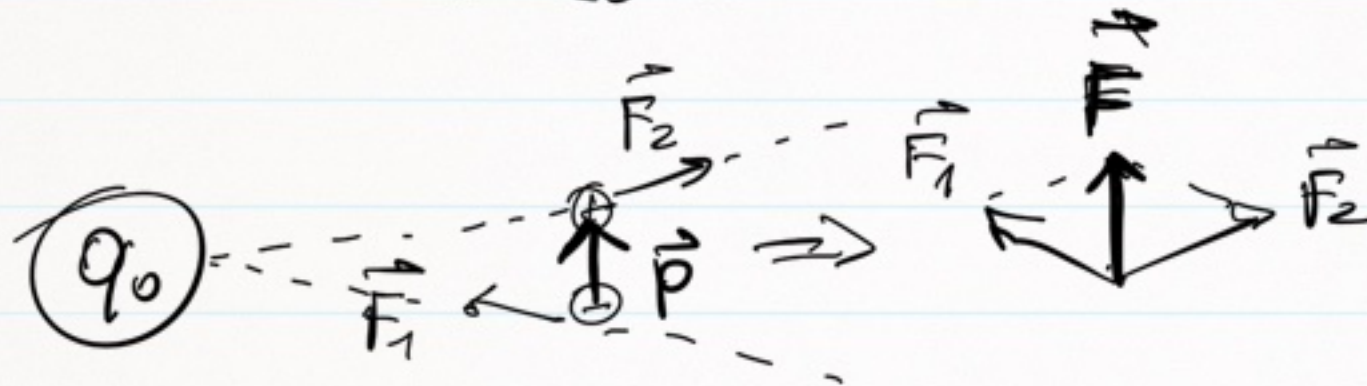
•  $\vec{p}(\theta, p, \theta) \Rightarrow \vec{p} \parallel \vec{E}$

$$F_y = \frac{p}{4\pi\epsilon_0} \cdot \frac{(-2z^2)}{r^5} \Rightarrow \vec{F} = - \frac{2pq_0}{4\pi\epsilon_0 r^3} \vec{u}_y$$

FORZA  
ATTRATTIVA

•  $\vec{p}(\theta, \theta, p)$

$$F_z = \frac{\rho q_0 z^2}{4\pi\epsilon_0 z^3} \Rightarrow \vec{F} = \frac{\rho q_0}{4\pi\epsilon_0 z^3} \vec{u}_z$$





N.B.: per campo elettrostatico, e quindi  
conservativo, vale:

$$\vec{F} = - \vec{\nabla} U_e = \vec{\nabla} (\vec{\rho} \cdot \vec{E}) \quad \text{ovvero}$$

$$F_i = \frac{\partial}{\partial i} (\vec{\rho} \cdot \vec{E}) = \rho_x \cdot \frac{\partial E_x}{\partial i} + \rho_y \cdot \frac{\partial E_y}{\partial i} + \rho_z \cdot \frac{\partial E_z}{\partial i}$$

con  $i = \{x, y, z\}$

Domanda ora visto a lezione, quando avete  
dimostrato Stokes, che valgono uguaglianze del tipo

$$\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}, \quad \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}, \quad \text{etc. etc.}$$